MP1994V2 Milestone Tracker

Mortensen–Pissarides (1994) Lean replication

1 August 2026

Current milestone: M9.2 – Statewise cutoffs, ordering, and regional formulas
Current status: NOT STARTED
Last completed milestone: M9.1 – Two-state foundations and coupled surplus system – **COMPLETE – GREEN**

Source of truth. `docs/milestone_tracker.md`. Status vocabulary: NOT STARTED; IN PROGRESS; BLOCKED; READY FOR REVIEW; COMPLETE – GREEN; COMPLETE – AMBER; DEFERRED / OUTSIDE CORE LEAN. M0 through M4 are complete and green. M5 is complete and amber: uniqueness is green and existence is conditional on an additional bracket. M5b is unstarted. M6 is complete and amber overall: its stock completion and uniqueness are green, while existence inherits M5's amber qualification. M7 is complete and green after human mathematical/economic review. M8 is COMPLETE – AMBER after review on 2026-08-01: analytic infrastructure is GREEN and supplied differentiable local paths cause the AMBER qualification. This is not inherited from M5. That historical grade applies to the original supplied-path layer. M8b.1 and M8b.2 derive all four paths from a supplied regular reduced equilibrium, so the full Appendix package is COMPLETE – GREEN under explicit Appendix matching/IFT regularity and sign conditions. Primitive selection of the base equilibrium would separately inherit M5's AMBER existence qualification.

Milestone plan

The tracker uses one readable block per milestone instead of a dense landscape table. Each block contains the same fields as the Markdown table.

M0. Foundations and primitive value equilibrium
Status: COMPLETE – GREEN
Paper: Sections 2–3; equations (1)–(7)
Economic target: Primitives, layered assumptions, probability/matching definitions, and primitive value equilibrium.
Expected declarations: `Primitives`; four assumption bundles; `cdf`; meeting rates; `surplus`; `continuationTerm`; `ValueEquilibrium`
Adequacy grade: GREEN **Dependencies:** None
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed M0 interface while beginning M1 separately.

M1. Surplus Bellman equation
Status: COMPLETE – GREEN
Paper: Section 3, journal p. 401; equation (8)
Economic target: Derive equation (8) from equations (3)–(7).
Expected declarations: Continuation decomposition; `firm_share`; `surplus_flow_equation`; `surplus_bellman_of_probability`; `surplus_bellman`
Adequacy grade: GREEN **Dependencies:** M0 complete
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed conditional representation theorem.

M2. Affine surplus and endogenous reservation cutoff
Status: COMPLETE – GREEN
Paper: Section 3; reservation discussion and equation (12)
Economic target: Surplus difference theorem, strict monotonicity, unique endogenous reservation threshold, and equation (12).
Expected declarations: Surplus difference, affinity, strict monotonicity, zero uniqueness, reservation theorem
Adequacy grade: GREEN **Dependencies:** M1
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-29 human mathematical/economic review; preserve M2 interfaces and design the measure-form continuation identity for M3.

M3. Continuation value, job destruction, and job creation
Status: COMPLETE – GREEN
Paper: Section 3; equations (9), (10), and (13)
Economic target: Measure-form continuation identity and derivation of job destruction and job creation.
Expected declarations: `expectedExcess`; `tailOptionValue`; equations (9), (10), and (13); JD/JC residual predicates; M3 capstone
Adequacy grade: GREEN **Dependencies:** M2
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-30 human mathematical/economic review; preserve the independent JD/JC interfaces and neutral capstone module.

M4. Reduced-equilibrium reconstruction and equivalence
Status: COMPLETE – GREEN
Paper: Section 3; equations (10), (13); journal p. 402
Economic target: Two-way representation between primitive values and the admissible reduced pair.
Expected declarations: `ReducedEquilibrium`; generic tail identity; forward and reverse maps; round trips; nonempty equivalence
Adequacy grade: GREEN **Dependencies:** M3
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-31 human mathematical/economic review; preserve representation interfaces and formulate corrected continuity, endpoint, and crossing assumptions for M5.

M5. Existence and uniqueness of static equilibrium
Status: COMPLETE – AMBER
Paper: Section 3; equations (10), (13), Figure 1, journal p. 402
Economic target: Existence and uniqueness of the static reduced equilibrium.
Expected declarations: Expected-excess regularity; JD/JC slopes; `StaticExistenceAssumptions`; conditional unique existence; value-side economic uniqueness
Adequacy grade: Uniqueness: GREEN; existence: AMBER; overall: AMBER **Dependencies:** M4
Blocker: Existence uses a lower positive crossing condition not derived from primitives
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-31 human review; preserve the conditional theorem and see the M5b design in `docs/static_existence_foundation.md`. This does not block M6.

M5b. Primitive foundation for static existence
Status: NOT STARTED
Paper: Foundational refinement to Section 3 static existence
Economic target: Replace the assumed lower crossing with a derived result.
Expected declarations: Proposed matching-boundary bundle; upper-job profitability; derived `lower_crossing`; optional Cobb–Douglas certification
Adequacy grade: Target: existence GREEN under a strengthened general theorem or certified Cobb–Douglas specialization **Dependencies:** M5
Blocker: The Inada condition on q alone is insufficient; JD-implied tightness must become positive through a separate profitability condition
Branch/commit or tag; last review; next action: No branch/tag or review; add right-hand Inada and upper-job profitability, derive `StaticExistenceAssumptions`, and see `docs/static_existence_foundation.md`. M5b is not M6.

M6. Unemployment flow balance and full static steady state
Status: COMPLETE – AMBER
Paper: Section 3; equation (14), Figure 2, journal pp. 403–404
Economic target: Derive unemployment, employment, vacancies, and the full static state.
Expected declarations: Strict mass/CDF bridge; hazards and flows; `steadyStateUnemployment`; `SteadyStateEquilibrium`; equation (14); exact reduced/full equivalence
Adequacy grade: Stock completion/equation (14): GREEN; uniqueness: GREEN; existence: AMBER inherited from M5; overall: AMBER **Dependencies:** M4–M5
Blocker: No new M6 blocker. Full-state nonemptiness inherits M5's unproved lower-crossing foundation
Branch/commit or tag; last review; next action: `issue2`; M6 completion commit; 2026-07-31 human mathematical/economic review; preserve the split capstones and adequacy contract; M7 is NOT STARTED.

M7. Static order comparative statics
Status: COMPLETE – GREEN
Paper: Section 3, journal pp. 401–404; equations (10), (13), (14); Figures 1–2
Economic target: Global order results for supplied equilibria, plus hazard, initial-flow, and stock implications for aggregate net productivity.
Expected declarations: Parameter updates/transports; four general fixed-tightness orders and two strict positive-option-value refinements; four robust equilibrium orders; hazard/flow/stock lemmas; two M7 capstones
Adequacy grade: GREEN **Dependencies:** M5–M6
Blocker: None for the proved pairwise results
Branch/commit or tag; last review; next action: `issue2`; M7 completion commit; reviewed 2026-07-31 by human mathematical/economic review; preserve the M7 interface and design M8's Appendix-specific derivative layer.

M8. Appendix derivative comparative statics
Status: COMPLETE – AMBER
Paper: Appendix; equations (A1)–(A12), plus equation (11)
Economic target: Derivative identities and signs along supplied differentiable equilibrium paths.
Expected declarations: Appendix matching assumptions; CDF/expected-excess analytic lemmas; exact path structures; (11), (A1)–(A12); component and overall capstones
Adequacy grade: Historical supplied-path layer AMBER; analytic statements remain valid **Dependencies:** M3–M4, M7 analytic interfaces
Blocker: Paths were supplied in M8; M8b now discharges that premise locally
Branch/commit or tag; last review; next action: `issue2`; M8 completion commit; reviewed 2026-08-01; preserve the historical record and use the GREEN M8b package for supplied regular equilibria.

M8b.1. Local IFT foundation and lambda path
Status: COMPLETE – GREEN
Paper: Appendix foundation
Economic target: Derive fixed-tightness sigma and lambda paths through a supplied reduced equilibrium.
Expected declarations: Appendix IFT regularity; parameterized residuals; scalar IFT wrapper; two path constructors and capstones
Adequacy grade: GREEN relative to a supplied equilibrium under explicit primitive C1 regularity of (q)
Dependencies: M8
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; reviewed 2026-08-01; preserve these interfaces as the foundation for completed M8b.2.

M8b.2. Discount and dispersion IFT paths
Status: COMPLETE – GREEN
Paper: Appendix foundation
Economic target: Construct discount and dispersion paths through a supplied regular equilibrium.
Expected declarations: Local C1 residuals; derived negative cutoff partials; two path constructors and capstones; full implicit Appendix capstone
Adequacy grade: Full Appendix package GREEN relative to a supplied regular reduced equilibrium **Dependencies:** M8b.1
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; reviewed 2026-08-02; begin M9 separately with aggregate-state transition terms.

M9.1. Anticipated two-state foundations
Status: COMPLETE – GREEN
Paper: Section 4; coupled system and conditional equations (16)–(18)
Economic target: Two-state primitives, primitive value equilibrium, and general max-based surplus system.
Expected declarations: Aggregate state; two-state primitives and values; coupled/state Bellman equations; conditional regional equations; capstone
Adequacy grade: GREEN **Dependencies:** M0–M3
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; reviewed 2026-08-02; preserve the conditional max-form representation.

M9.2. Cutoffs, ordering, and certified regions
Status: NOT STARTED
Paper: Section 4; cutoff-ordering discussion and regional formulas
Economic target: Statewise cutoffs, their economic admissibility and ordering, and certified interval formulas.
Expected declarations: Statewise cutoff definitions; monotonicity/affinity; admissibility; ordering theorem; regional package
Adequacy grade: – **Dependencies:** M9.1
Blocker: Ordering must be proved from the coupled system
Branch/commit or tag; last review; next action: No branch/tag or review; prove ordering before simplifying positive parts by interval.

M9.3. Job creation, dynamics, and impact asymmetry
Status: NOT STARTED
Paper: Section 4; remaining equations (15)–(31)
Economic target: Statewise job creation, unemployment dynamics, and impact asymmetry.
Expected declarations: Two-state JC; employment-measure dynamics; impact flow accounting and asymmetry capstone
Adequacy grade: – **Dependencies:** M9.2, M6
Blocker: Employment-measure transition interface
Branch/commit or tag; last review; next action: No branch/tag or review; begin only after M9.2 certifies cutoff regions.

M10. Finite Markov equilibrium and employment accounting
Status: NOT STARTED
Paper: Analytical prelude to Section 5; equations (32)–(38)
Economic target: Finite-state equilibrium representation, employment transition operator, creation/destruction accounting, and equation (38).
Expected declarations: Finite kernel; transition operator; mass preservation; accounting identity
Adequacy grade: – **Dependencies:** M6, M9
Blocker: Finite-state interface and numerical-solver connection
Branch/commit or tag; last review; next action: No branch/tag or review; prefer finite state before general measurable states.

NUM. Numerical simulation and calibration
Status: DEFERRED / OUTSIDE CORE LEAN
Paper: Section 5; equations (39)–(42), Tables I–II
Economic target: External Python or Julia numerical replication; Lean only for exact accounting or certified residuals where useful.
Expected declarations: External solver and optional Lean residual/accounting certificates
Adequacy grade: – **Dependencies:** Analytical interfaces as needed
Blocker: Numerical/empirical work is outside core Lean
Branch/commit or tag; last review; next action: No branch/tag or review; define a reproducible calibration and residual protocol externally.

Adequacy notes. M6: `docs/static_steady_state_adequacy.md`. Inherited M5 existence foundation: `docs/static_existence_foundation.md`. M7 scope: `docs/static_comparative_statics_scope.md`. M8 scope: `docs/appendix_differentiability_scope.md`.

M0 acceptance checklist

- ☒ New module root is isolated from the legacy architecture.
- ☒ Primitive parameter structure exists.
- ☒ Economic assumptions are layered.
- ☒ Shock law is represented by a measure.
- ☒ CDF is derived from the measure.
- ☒ Matching rates are defined.
- ☒ Surplus is defined from J, W, U, not stored independently.
- ☒ Equations (1)–(7) are represented faithfully.
- ☒ Equations (5) and (6) use one shared continuation term.
- ☒ No cutoff or reduced-equilibrium conclusion is assumed.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No `sorry`, `admit`, or custom axioms exist.
- ☒ Tracker TeX compiles and PDF has been visually inspected.
- ☒ Human mathematical/economic review is complete.

M1 acceptance checklist

- ☒ Generic continuation decomposition is proved under a probability measure.
- ☒ Firm surplus share is derived from equations (3) and (4).
- ☒ The surplus flow equation is derived from equations (5)–(7).
- ☒ Equation (8) is proved for the actual equilibrium surplus.
- ☒ The main theorem uses only probability normalization and existing integrability.
- ☒ The `ShockAssumptions` wrapper installs the probability instance locally.
- ☒ No M0 structure was strengthened or modified.
- ☒ No cutoff, affine surplus, or reduced equilibrium was introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No `sorry`, `admit`, or custom axioms exist.
- ☒ Proof ledger TeX/PDF compiles and passes visual inspection.
- ☒ Milestone tracker TeX/PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

M2 acceptance checklist

- ☒ Equation (8) is subtracted at two states.
- ☒ The scaled two-point surplus identity is proved.
- ☒ The divided affine-surplus identity is proved.
- ☒ Surplus is proved strictly increasing.
- ☒ The cutoff is defined from actual equilibrium surplus.
- ☒ The cutoff is proved to be a surplus zero.
- ☒ The zero is proved unique.
- ☒ $V = 0$ is derived from free entry and $r > 0$.
- ☒ $q(\theta)J(\varepsilon_u) = c$ follows from (1)–(2).
- ☒ $J(\varepsilon_u) > 0$ is proved.
- ☒ $S(\varepsilon_u) > 0$ is proved.
- ☒ $\varepsilon_d < \varepsilon_u$ is proved.
- ☒ Equation (12) is proved.
- ☒ Surplus signs are characterized relative to ε_d.
- ☒ Firm-value signs are characterized relative to ε_d.
- ☒ Active surplus is rewritten as a scaled positive part.
- ☒ No M0 or M1 mathematical declaration was changed.
- ☒ No M3 or later theorem was introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No `sorry`, `admit`, or custom axioms exist.
- ☒ Proof-ledger PDF compiles and passes visual inspection.
- ☒ Tracker PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

M3 acceptance checklist

- ☒ Expected excess is the integral of the positive part.
- ☒ Tail option value is defined from the induced CDF.
- ☒ Expected-excess integrability is proved.
- ☒ Strict upper-tail probability equals one minus the CDF.
- ☒ The strict-tail layer-cake identity is proved.
- ☒ Active-surplus expectation is rewritten using expected excess.
- ☒ Measure-form equation (9) is derived.
- ☒ Paper-facing equation (9) is derived.
- ☒ The free-entry search-gain identity is derived.
- ☒ Measure-form equation (10) is derived.
- ☒ Paper-facing equation (10) is derived.
- ☒ Job-destruction residual and predicate are defined.
- ☒ The equilibrium pair satisfies the job-destruction predicate.
- ☒ The product-form job-creation identity is derived.
- ☒ Paper equation (13) is derived.
- ☒ Job-creation residual and predicate are defined.
- ☒ The equilibrium pair satisfies the job-creation predicate.
- ☒ No `ReducedEquilibrium` structure is introduced.
- ☒ No M0–M2 mathematical declaration is changed.
- ☒ No M4 or later theorem is introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No `sorry`, `admit`, custom axiom, or opaque placeholder exists.
- ☒ Proof-ledger PDF compiles and passes visual inspection.
- ☒ Tracker PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

M4 acceptance checklist

☒ Robust product-form job-creation predicate is defined.
☒ Reduced pair stores positive theta and an admissible cutoff.
☒ Measure-form JD and product-form JC are the only stored equations.
☒ Generic positive-part integrability and tail identity are proved.
☒ Reduced paper equations (10) and (13) are derived.
☒ Every value equilibrium maps to a reduced equilibrium.
☒ Candidate surplus, J, U, W, V, and explicit Nash wage are defined.
☒ Active surplus is integrable and candidate equation (8) is proved.
☒ Reconstructed equations (1)–(7) are proved.
☒ A reduced pair reconstructs a value equilibrium and its cutoff.
☒ Reconstruct–reduce is exact; reverse economic components agree.
☒ Conditional nonempty equivalence is proved.
☒ No unconditional existence, uniqueness, M0–M3 change, or M5 theorem.
☒ Targeted and full builds pass; audits and PDFs pass.
☒ Human mathematical/economic review is complete.

M5 acceptance checklist

☒ Expected excess is antitone, one-Lipschitz, and continuous.
☒ JD net value and JD-implied tightness are defined.
☒ JD satisfaction is equivalent to $\theta = \theta_{JD}(d)$.
☒ The JD curve is strictly increasing.
☒ The scalar JC residual is defined and strictly decreasing on its admissible domain.
☒ Figure 1 JD-upward and JC-downward slope theorems are proved.
☒ At most one `ReducedEquilibrium` is proved.
☒ `StaticExistenceAssumptions` contains no root or equilibrium witness.
☒ Bracket continuity and upper residual $-c$ are proved.
☒ An interior root and a reduced witness are constructed by the IVT.
☒ Conditional unique reduced existence is proved.
☒ Value nonemptiness, theta/cutoff uniqueness, and economic uniqueness follow through M4.
☒ Existence is labeled conditional on the additional bracket assumptions.
☒ No M0–M4 theorem statement changed and no M6 theorem was introduced.
☒ Targeted/full builds, audits, and both PDFs pass.
☒ Human mathematical/economic review is complete.

M6 acceptance checklist

- ☒ Strict destruction mass matches the M2 rule and equals the CDF under no atoms.
- ☒ Separation/job-finding hazards, creation/destruction flows, and the residual are defined.
- ☒ Total transition hazard is positive.
- ☒ Closed-form unemployment is in $[0, 1)$, balances flows, and is unique.
- ☒ Paper equation (14) is proved.
- ☒ `SteadyStateEquilibrium` is defined; employment and vacancies are derived.
- ☒ Creation is $q(\theta)v$; destruction is $\lambda F(d)(1 - u)$; flows are equal.
- ☒ Positive separation implies positive unemployment and vacancies.
- ☒ $v/u = \theta$ is proved only under positivity.
- ☒ Every reduced equilibrium completes to a full state; exact round trips and an equivalence are proved.
- ☒ At most one full state is proved without existence assumptions.
- ☒ Conditional nonemptiness and unique existence are transported from M5.
- ☒ No Beveridge-curve slope, M0–M5 statement change, or M7+ theorem was introduced.
- ☒ Targeted/full builds, no-sorry/axiom audits, and prohibited-pattern searches pass.
- ☒ Proof-ledger and tracker PDFs compile and pass visual inspection.
- ☒ Human mathematical/economic review is complete.

M7 acceptance checklist

- ☒ Parameter-update functions are defined and assumption transport is proved.
- ☒ Fixed-tightness cutoff results for p, b, λ, r are proved.
- ☒ Full-equilibrium p and b orders are proved.
- ☒ Full-equilibrium λ cutoff and r tightness orders are proved.
- ☒ Separation-hazard and job-finding-rate orders are proved.
- ☒ Initial creation/destruction flow results are proved.
- ☒ Steady unemployment and employment orders are proved.
- ☒ The vacancy response is explicitly left ambiguous.
- ☒ No core theorem uses `StaticExistenceAssumptions`.
- ☒ M8 derivative results remain unimplemented.
- ☒ No M0–M6 mathematical theorem changed.
- ☒ Targeted and full builds pass.
- ☒ No `sorry`, `admit`, custom axiom, or opaque placeholder exists.
- ☒ Proof-ledger and tracker PDFs pass visual inspection.
- ☒ Human mathematical/economic review is complete.

M8 acceptance checklist

- ☒ Appendix assumptions are isolated from the core.
- ☒ Appendix matching elasticity and $0 < \eta < 1$ are explicit.
- ☒ CDF continuity and the expected-excess derivative are derived from the shock law.
- ☒ Weak/strict tail-gap bounds and moment identities use explicit shock assumptions.
- ☒ Path structures contain exact differentiability and closure, but no sign conclusion.
- ☒ Equation (11), its sign characterization, and (A1)–(A12) are proved.
- ☒ Lambda cutoff and tightness derivatives are negative.
- ☒ Discount tightness is negative and the cutoff has the exact ambiguous (A8) condition.
- ☒ Dispersion derivatives are positive under $b \leq p$.
- ☒ The lambda capstone explicitly takes $0 < \lambda$.
- ☒ No result takes `StaticExistenceAssumptions` or selects an M5 witness.
- ☒ M8's AMBER source is supplied-path regularity, not M5 inheritance.
- ☒ The six-step M8b route is documented; no M0–M7 theorem changed.
- ☒ The M8 review surface was isolated from later M8b and Section 4 work.
- ☒ All ten M8 sources pass direct elaboration; aggregate and full builds pass.
- ☒ No sorry, admit, custom axiom, or opaque placeholder exists.
- ☒ Transitive axiom checks contain no custom project axiom.
- ☒ Both PDFs pass visual inspection and the review archive excludes build artifacts.
- ☒ Human mathematical/economic review is complete (2026-08-01).

M8b.1 acceptance checklist

- ☒ Appendix IFT assumptions contain only local C1 regularity of (q).
- ☒ The installed Mathlib IFT API is recorded.
- ☒ Expected-excess local C1 regularity is derived.
- ☒ Both base roots and cutoff nondegeneracy results are proved.
- ☒ The generic wrapper exposes local closure and graph uniqueness.
- ☒ Fixed-sigma and lambda paths and their M8 capstones are constructed.
- ☒ No M5-selected equilibrium or static-existence assumption is used.
- ☒ The reviewed M8b.1 layer contains no M8b.2 or Section 4 theorem.
- ☒ Builds and source audits pass; PDFs are visually inspected.
- ☒ Human mathematical/economic review is complete (2026-08-01).

M8b.2 acceptance checklist

- ☒ Discount and dispersion base roots are derived from the supplied equilibrium.
- ☒ Both cutoff partials are proved strictly negative and nonzero.
- ☒ Joint local C1 regularity is proved for both JD graphs and residuals.
- ☒ Both equilibrium paths are constructed by the scalar IFT.
- ☒ Local positivity, JD, product-form JC, and base recovery are proved.
- ☒ Discount preserves the exact A8 iff; dispersion signs use normalization and $b \leq p$.
- ☒ The full capstone takes no path, M5 selection, sign, or nondegeneracy assumption.
- ☒ No Section 4 declaration is implemented.
- ☒ Direct, aggregate, audit, and full builds pass.
- ☒ Source audits and PDF visual inspection pass.
- ☒ Human mathematical/economic review is complete (2026-08-02).

M9.1 acceptance checklist

- ☒ `AggregateState` is defined.
- ☒ State switching is involutive.
- ☒ Two-state productivity primitives are defined.
- ☒ High productivity and positive transition rate are primitive assumptions.
- ☒ No cutoff ordering is assumed.
- ☒ Two-state value candidates are defined.
- ☒ Statewise surplus is derived from (J,W,U).
- ☒ Statewise active-surplus integrability is required.
- ☒ Vacancy and unemployment equations include aggregate capital gains.
- ☒ Firm and worker aggregate continuation terms are economically correct.
- ☒ Firm share and zero vacancy values are derived.
- ☒ Raw and general coupled max-form surplus equations are derived.
- ☒ Recession and boom max-form equations are derived.
- ☒ Conditional equations (16), (17), and (18) are derived.
- ☒ No cutoff or cutoff ordering is introduced.
- ☒ No equation (15) or equations (19)–(30) are introduced.
- ☒ No M0–M8b.2 theorem changes.
- ☒ Targeted, aggregate, audit, and full builds pass.
- ☒ No placeholder or custom axiom exists.
- ☒ Both PDFs pass visual inspection.
- ☒ Human mathematical/economic review is complete (2026-08-02).

M5b foundational follow-up

Status: NOT STARTED. The adequacy target is to upgrade static existence from amber to green under either a strengthened general matching-boundary theorem or a fully certified Cobb–Douglas specialization. M5b is a foundational refinement, not M6, and is not part of the current M5 Lean proof.

The proposed work adds a right-hand Inada condition for q, separately requires $b < p + \sigma\varepsilon_u$, proves that the JD curve reaches zero and then positive tightness, derives `lower_crossing`, and constructs `StaticExistenceAssumptions`. The Inada condition on q alone is insufficient because it says nothing about whether JD-implied tightness ever becomes positive. The complete design is in `docs/static_existence_foundation.md`. M6 does not implement this refinement.

Changelog

2026-08-02 – M9.1 human review completed. Review confirmed a conditional representation theorem for every supplied two-state value equilibrium; existence is not asserted and does not make the result AMBER. Aggregate capital gains are present in all four primitive value equations, and the coupled surplus equation is derived. Equations (16)–(18) are max-form sign reductions rather than final cutoff-indexed interval formulas. No cutoff or ordering is assumed. M9.1 is **COMPLETE – GREEN**; M9.2 is next and NOT STARTED.

2026-08-02 – M9.1 anticipated two-state foundations implemented. Added six cyclical modules for the recession/boom process, primitive value system, shared continuation terms, general coupled max-based surplus equation, conditional equations (16)–(18), and capstone. Probability remains explicit through `ShockAssumptions`; all results are conditional on a supplied `TwoStateValueEquilibrium`. No cutoff, ordering, existence, job-creation, unemployment-dynamics, or impact-asymmetry theorem was introduced. M9.1 is **READY FOR REVIEW**, target GREEN; M9.2 and M9.3 remain NOT STARTED.

2026-08-02 – M8b.2 human review completed. The discount residual/path, dispersion residual/path, and full IFT-derived Appendix capstone are **COMPLETE – GREEN**. Roots and strictly negative cutoff partials are derived, all four paths are constructed through Mathlib's IFT, and no path, sign, M5 selection, or static-existence assumption is used. The full Appendix package is GREEN relative to a supplied regular reduced equilibrium and explicit Appendix regularity/sign conditions. Primitive base selection would separately inherit M5 AMBER. M9 is next and remains **NOT STARTED**.

2026-08-01 – M8b.2 discount and dispersion paths implemented. Derived both base roots and strictly negative cutoff partials from a supplied reduced equilibrium, proved joint local C1 regularity, constructed both local equilibrium paths by the scalar IFT, and connected them to the reviewed M8 discount and dispersion results. The full implicit Appendix capstone has no M5, supplied-path, sign, or nondegeneracy premise. M8b.2 is **READY FOR REVIEW**, target GREEN; overall M8 remains **COMPLETE – AMBER** pending review.

2026-08-01 – M8b.1 human review completed. The generic wrapper, fixed-sigma path, lambda path, and combined result are **COMPLETE – GREEN** relative to a supplied equilibrium under explicit Appendix IFT regularity. Base roots and nondegeneracy are proved; no sign or M5 selection is assumed. The C1 condition on (q) is a technical strengthening, not a paper conclusion, and no global uniqueness is claimed. Overall M8 stays **COMPLETE – AMBER** pending M8b.2 review.

2026-08-01 – M8b.1 local IFT foundation implemented. Added primitive matching C1 regularity, parameterized residual calculus, the generic scalar Mathlib IFT wrapper, and derived fixed-tightness sigma and lambda equilibrium paths. Base roots and nonzero cutoff derivatives are proved; none is assumed. No M5 existence result is used. M8b.1 is **READY FOR REVIEW**, target GREEN; M8 remains **COMPLETE – AMBER** until M8b.2.

2026-08-01 – M8 human review completed. The Appendix identities and signs were found faithful and non-circular. Analytic infrastructure is **COMPLETE – GREEN**; supplied-path results and overall M8 are **COMPLETE – AMBER**. The AMBER source is supplied differentiable local paths, not M5. The lambda capstone uses $0 < \lambda$, and M8b remains the route to GREEN for the path layer.

2026-08-01 – M8 Appendix derivative comparative statics integrated. Added isolated Appendix assumptions, shock-law analytic lemmas, exact supplied paths, scalar algebra, equations (11) and (A1)–(A12), sign theorems, and capstones. Lambda lowers cutoff and tightness; discount lowers tightness and has the exact (A8) cutoff condition; dispersion raises both derivatives under $b \leq p$. No M8 theorem uses `StaticExistenceAssumptions` or an M5 witness. The missing IFT path construction is the sole AMBER gap and is deferred to M8b. M8 is **READY FOR REVIEW**, overall **COMPLETE – AMBER**; human review remains pending.

2026-07-31 – M7 human review completed. The order proofs were found mathematically and economically sound; M7 is **COMPLETE – GREEN**. No M5 existence assumption is used. Unnecessary imports of the M5 existence and M6 transported-existence layers were removed. Strict fixed-tightness refinements were added under explicit positive option value, with positive λ additionally required for the strict discount-rate result. Proof-ledger symbol and rendering defects were corrected. The λ-tightness, r-cutoff, σ, and derivative results remain M8, which is NOT STARTED.

2026-07-31 – M7 static order comparative statics implemented. Added immutable parameter updates and transports, four fixed-tightness cutoff orders, robust supplied-equilibrium orders for p, b, λ, r, and weak hazard, flow, unemployment, and employment implications for aggregate net productivity. Vacancies remain ambiguous. No M7 core result uses `StaticExistenceAssumptions`; existence is not asserted. The λ-tightness, r-cutoff, σ, and Appendix derivative results remain M8. Added `docs/static_comparative_statics_scope.md`. M7 is **READY FOR REVIEW**, target **COMPLETE – GREEN**.

2026-07-31 – M6 human review completed. Stock completion, equation (14), accounting, exact representation, and full-state uniqueness were judged mathematically and economically faithful and are GREEN. Zero separation correctly yields $u = v = 0$, with the literal ratio guarded by positive separation. M6 adds no existence closure premise; existence inherits M5's `lower_crossing` and remains AMBER. Added split green/amber capstones and the detailed record `docs/static_steady_state_adequacy.md`. M6 is **COMPLETE – AMBER**; M7 is **NOT STARTED**.

2026-07-31 – M6 unemployment flow balance implemented. Added the strict destruction mass and atomless CDF bridge, hazards, flows, the unique balanced unemployment formula, and equation (14). Added the full static state with derived employment/vacancies, guarded $v/u = \theta$ by positive separation, and proved exact reduced/full equivalence, uniqueness, and conditional existence inherited from M5. Stock completion is recommended GREEN; existence and overall M6 were submitted as AMBER pending the completed review above.

2026-07-31 – M5 static-existence foundation documented. Recorded the exact current continuity and lower-crossing assumptions and the unimplemented M5b route combining right-hand Inada behavior with separate upper-job profitability. The note makes explicit that Inada alone is insufficient. M5 remains **COMPLETE – AMBER**; M6 remains **NOT STARTED**. The durable design note is `docs/static_existence_foundation.md`.

2026-07-31 – M5 review completed. The uniqueness and IVT proofs passed review. Figure 1 establishes uniqueness, not existence. The non-circular `lower_crossing` field assumes a positive residual rather than a root, but supplies the missing endpoint condition. Existence-only signatures now omit matching assumptions. Uniqueness is **COMPLETE – GREEN**; existence and overall M5 are **COMPLETE – AMBER**. A future task is to derive the lower crossing from primitive conditions on q, or certify it for Cobb–Douglas; this does not block M6.

2026-07-31 – M5 static existence and uniqueness implemented. Expected excess is one-Lipschitz; JD slopes upward and JC downward, proving at most one reduced equilibrium. The separate continuity/lower-bracket bundle and the derived upper residual $-c < 0$ yield an interior crossing by the IVT. M4 transports conditional existence and economic uniqueness to value equilibria. The implementation was submitted for review; equation (14) remains M6.

2026-07-31 – M4 review completed. Review found the generic excess and CDF-tail analysis valid, and equations (10) and (13) sufficient for reconstruction under the stated admissibility assumptions. The Nash wage is economically correct; equations (1)–(7) are reconstructed; and both economic round trips are certified. The import diagram and assumption-field account were corrected. M4 proves no existence or uniqueness theorem and is **COMPLETE – GREEN**.

2026-07-30 – M4 reconstruction and equivalence implemented. Created the robust reduced pair, generic analytic layer, forward bridge, explicit reverse reconstruction, and equivalence module. Equations (1)–(7) are reconstructed with the Nash wage; the reduced round trip is exact and the value round trip is componentwise exact. Nonemptiness equivalence consumes a witness on either side and proves no existence or uniqueness. M4 is **READY FOR REVIEW**.

2026-07-30 – M3 review completed. The layer-cake/CDF proof was reviewed and found valid without atomlessness. Equations (9), (10), and (13) have the correct signs and dependencies, and job creation is independent of the job-destruction equation. The capstone was moved to the neutral combining module `SteadyState/StaticConditions.lean`. Residual predicates require separate admissibility fields in M4, which must also prove a generic expected-excess/tail theorem for reverse reconstruction. M3 was graded **COMPLETE – GREEN**.

2026-07-30 – M3 continuation and static conditions implemented. Created `SteadyState/Continuation.lean`, `SteadyState/JobDestruction.lean`, and `SteadyState/JobCreation.lean`, with the capstone in the neutral `SteadyState/StaticConditions.lean`. Expected excess follows from the M2 active-surplus formula; Mathlib's strict-tail layer-cake theorem gives the CDF representation without atomlessness. Equations (9), (10), and (13), the two residual predicates, and the forward M3 capstone are proved for an existing `ValueEquilibrium`. No reduced-equilibrium structure or M4 theorem was introduced.

2026-07-29 – M2 review completed. Review confirmed that equation (8) was correctly subtracted at two states and that actual equilibrium surplus is affine and strictly increasing. The derived `reservationCutoff` is its uniquely identified zero; equations (1)–(2) establish $\varepsilon_d < \varepsilon_u$; and equation (12) and the reservation rule are proved. M2 remains conditional on an existing `ValueEquilibrium` and was graded **COMPLETE – GREEN**.

2026-07-29 – M2 affine surplus and cutoff implemented. Created `SteadyState/AffineSurplus.lean` and `SteadyState/Cutoff.lean`. Subtracting equation (8) gives affine surplus and strict monotonicity. The derived cutoff is the unique surplus zero; equations (1)–(2), positive vacancy cost and meeting, and $\beta < 1$ separately place it below ε_u. Added equation (12), reservation sign rules, the active-surplus bridge, and the M2 capstone. M2 is **READY FOR REVIEW**.

Review-archive convention. Beginning with M3, project-local review archives are named `review_m3.zip`, `review_m4.zip`, and so on. They remain untracked and are never committed.

2026-07-29 – M1 review completed. Human mathematical and economic review found equation (8) mathematically faithful to the paper and its proof non-circular. It is derived from equations (3)–(7), uses the actual equilibrium surplus, and assumes no cutoff, affinity result, or reduced equilibrium. It is a conditional representation theorem for an existing `ValueEquilibrium`; M1 does not prove equilibrium existence. M1 was graded **COMPLETE – GREEN**.

2026-07-29 – M1 equation (8) implemented. Created `MP1994V2/SteadyState/Surplus.lean` with the continuation decomposition, firm share, surplus flow equation, minimal probability theorem, and `ShockAssumptions` wrapper. Updated `All`, `Audit`, `README`, trackers, and added the proof ledger. M1 is **READY FOR REVIEW**.

2026-07-29 – M0 review completed. Review found equations (1)–(7) economically faithful and no future conclusion embedded; corrected substantive modules → `All` → `Audit`; required explicit `ShockAssumptions P` and local `D.isProbability` in M1; graded **COMPLETE – GREEN**.

2026-07-29 – v2 foundation created. Added `Definitions`, `Assumptions`, `Probability`, `Matching`, `Equilibrium/Value`, `All`, `Audit`, `README`, and synchronized trackers: primitives, four assumption bundles, CDF/support/matching notation, shared surplus/continuation, `ValueCandidate`, and `ValueEquilibrium`. No legacy v1 Lean or architecture file changed; the library-root import makes `lake build` certify `MP1994V2.All`.